

Pythagoras' Theorem & Trigonometry

Extra practice test 2 — answer sheet · 27 marks total

Section A — Pythagoras' theorem

A1. $a^2 + b^2 = 9^2 + 12^2 = 81 + 144 = 225$ (1) $c = \sqrt{225} = \mathbf{15\text{ cm}}$ (1)

A2. $a^2 = 17^2 - 8^2$ (1) $= 289 - 64 = 225$ (1) $a = \sqrt{225} = \mathbf{15\text{ cm}}$ (1)

A3. $\text{diagonal}^2 = 9^2 + 40^2 = 81 + 1600 = 1681$ (1) $\text{diagonal} = \sqrt{1681} = \mathbf{41\text{ cm}}$ (1)

Section B — SOH CAH TOA basics

B1. The **hypotenuse** (1)

B2. **CAH**: $\text{adjacent} = \text{hypotenuse} \times \cos \theta$ (1)

B3. **SOH**: $\theta = \sin^{-1}(\text{opposite} \div \text{hypotenuse})$ (1)

Section C — Finding missing sides

C1. $\sin 37^\circ = x \div 12$ (1) $x = 12 \times \sin 37^\circ$ (1) $= \mathbf{7.2\text{ cm}}$ (1)

C2. $\cos 62^\circ = 5 \div h \rightarrow h = 5 \div \cos 62^\circ$ (1) $= \mathbf{10.7\text{ cm}}$ (1)

C3. $\tan 27^\circ = x \div 9 \rightarrow x = 9 \times \tan 27^\circ$ (1) $= \mathbf{4.6\text{ cm}}$ (1)

Section D — Finding missing angles

D1. $\sin \theta = 8 \div 17 = 0.4706 \rightarrow \theta = \sin^{-1}(0.4706)$ (1) $= \mathbf{28.1^\circ}$ (1)

D2. $\tan \theta = 11 \div 7 = 1.5714$ (1) $\theta = \tan^{-1}(1.5714)$ (1) $= \mathbf{57.5^\circ}$ (1)

Section E — Applying Pythagoras & trigonometry

E1. Use Pythagoras: $\text{height}^2 = 5.8^2 - 2.1^2 = 33.64 - 4.41 = 29.23$ (1) $\sqrt{29.23} = \mathbf{5.4\text{ m}}$ (1)

E2. $\tan 50^\circ = h \div 8$ (1) $h = 8 \times \tan 50^\circ$ (1) $= \mathbf{9.5\text{ m}}$ (1)

Marking note: award method marks for a correctly set-up equation or calculation even if the final answer contains a rounding or arithmetic slip. Accept any correct rounding to 1 decimal place unless a different level of accuracy is specified. Missing units on an otherwise-correct numerical answer should lose the final accuracy mark, not the method mark.